2017 EXAMINATIONS

Lancaster University

Part II

COMPUTING AND COMMUNICATIONS **[2.5 Hours]**

SCC.366 Media Coding and Processing

Candidates are asked to answer THREE questions from FOUR; each question is worth a total of 25 marks.

Use a separate answer book for each question.

Question 1

This question addresses the Data Coding.

An image coding scheme uses a symbol encoding algorithm that converts a sequence of input source symbols to a binary digit sequence.

Consider a source symbol generation process where the source symbols are given as a set with 6 elements: S = {A,B,C,D,E,F} and the probability of the occurrence of each symbol in S is given as:

P(B)	*P*(A)	*P*(F)	*P*(D)	*P*(E)	*P*(C)
0.35	0.40	0.03	0.07	0.05	0.10

1.a Determine the minimum number of bits required per code word when the input source symbol sequence is encoded with the fixed-length encoding (FLE) scheme. Show an example FLE code (Code1) of this case.

[3 marks]

1.b

i. Obtain the average information in this symbol set **[2 marks]**

ii. Design a Huffman code (Code2) for this source and show the Huffman tree. In the designed code, the symbol with the shortest code length should be assigned with the code word `0'.

[8 marks]

iii. Calculate the average number of bits per code word for Code2. **[2 marks]**

1.c Using Code2, provide the result of decoding the following code word sequence back to the original sequence of symbols in S={A,B,C,D,E,F}.

[111111001100011110111 00]

[5 mark]

1.d Provide the results of run-length encoding for the two source sequences that are given below. The outputs are given as sequences of symbols in S={A,B,C,D,E,F} and natural numbers.

[BBCCCCCCCCCCBBBBBAAAAADDDDDD]
[BCCCBCCBCCBCBBACCABADDDAADDD]

[5 marks]

Total 25 marks

Question 2

This question addresses the Image/video Coding.

2.a

i. Explain in detail how an audio signal is encoded and decoded using PCM, DPCM and ADPCM and compare their coding efficiency.

ii. In order to improve the compression ratio many audio compression schemes exploit masking effects; describe in detail the different types of masking found in audio and explain how masking is used to improve the level of compression.

[8 marks]

2.b A number of media standards, including H.263 and MPEG-2, have support for layered coding.

i. Outline a scenario that demonstrates the need for layered coding.

ii. Describe four types of scalable video provided by MPEG-2, and for each approach outline how scalability is implemented. You are encouraged to use diagrams to illustrate your answer.

[12 marks]

2.c MPEG-4 defines coding types for Rectangular Video and Arbitrarily Shaped Video. Explain the difference between these two concepts. What is the idea behind Arbitrarily Shaped Video and what concepts have been standardised to achieve this? Also describe how a scene can be created in the case of Arbitrarily Shaped Video.

[5 marks]

Total 25 marks

Question 3

This question addresses the JPEG encoding standard, in which an input image is digitised and coded.

3.a

i. Provide block diagrams that describe the encoding and decoding processes in JPEG. The encoding and decoding processes contain three and two main components (blocks), respectively. In this diagram, identify a specific component in the encoding step which is responsible for data loss.

[5 marks]

ii. Explain how the trade-off between compression ratio and reconstruction quality (i.e., data loss) is determined in JPEG

[2 marks]

3.b

i. The RGB colour values of the input image are converted to UC_bC_r colour values and optionally, selected component(s) among U, C_b, C_r is(are) resized into low-resolution images before the subsequent encoding process. Explain which component(s) can be resized. Explain your choice.

[5 marks]

ii. The discrete cosine transform (DCT) is used as a data mapper. In general, any linear mapper can be adopted instead of the DCT. Explain the pros and cons of using principal component analysis as a data mapper instead of DCT.

[5 marks]

iii. After the DCT step, a mapped block contains DC and AC components. Explain which components are encoded with differential pulse code modulation and/or run-length encoding? Explain your choice.

[5 marks]

3.c Consider the following numerical symbol generation and quantization processes: Source symbols are generated as integers in [0,255], i.e., S={0, 1, ..., 255}. All symbols in S are uniformly quantized into 10 equal-sized intervals. This leads to a quantized symbol set Q(S) of size 10. Provide the quantization results of input source sequence given below.

$$[254,16,80,174,202,101,62] \subset S$$
$$[?, ?, ?, ?, ?, ?, ?] \subset Q(S)$$

[3 marks]

Total 25 marks

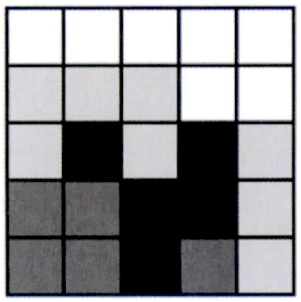
Question 4

This question addresses the Image/video processing.

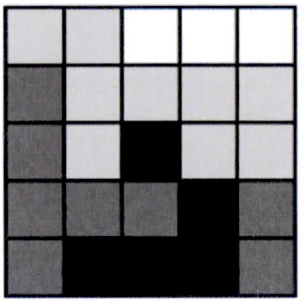
4.a An approach to conduct cut-detection is using histograms. However, there are instances where this fails. Briefly describe four different scenarios in which cut detection based on histograms typically fails and explain why. **[5 marks]**

4.b

(1)



(2)



The above figures show two image sections of size 5x5 pixels. The images are greyscale images with four different shades of grey. The shades are: 0 (black), 1 (dark grey), 2 (light grey), and 3 (white).

i. Compute the two greyscale histograms for the images 1 and 2.
ii. Compute the histogram difference between image 1 and image 2.

Show the process by which you arrived at your answer.

[8 marks]

4.c Describe the stages at which automatic video analysis process take place and explain what are these processes and what kind of information/features do they extract.

[6 marks]

4.d Describe format specific watermarking and explain its advantages and disadvantages (and list one advantage and one disadvantage).

[6 marks]

Total 25 marks

--- End of Paper ---